

Autoencoder anomaly detection · Taiwan Cancer Registry
78,442 patients · 37 sites · 7 syndrome patterns

Method: Bottleneck autoencoder (dim=16) trained on all patients.
Anomaly score = per-patient sum-BCE. Top 1% (784 patients) screened.
Match threshold ≥ 0.4 of syndrome-defining sites present.

Total candidates: 128

Syndrome definitions (effective sites in registry):

Lynch: ['C16', 'C18', 'C20', 'C54', 'C56', 'C67']

BRCA: ['C50', 'C56', 'C61']

MEN1: ['C18']

Li-Fraumeni: ['C50', 'C71']

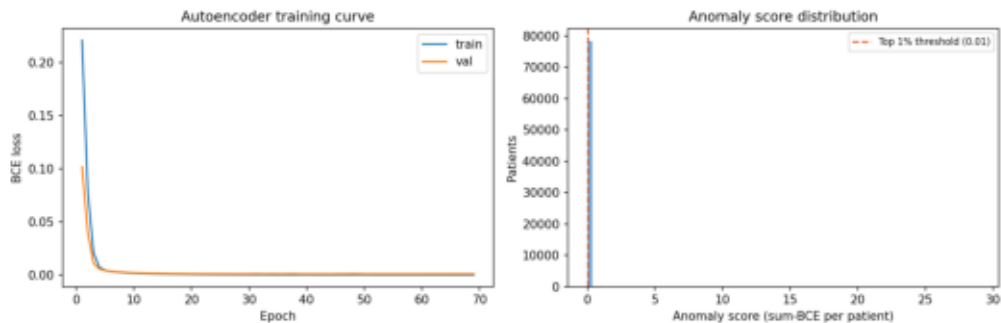
FAP: ['C16', 'C18', 'C20']

VHL: ['C71']

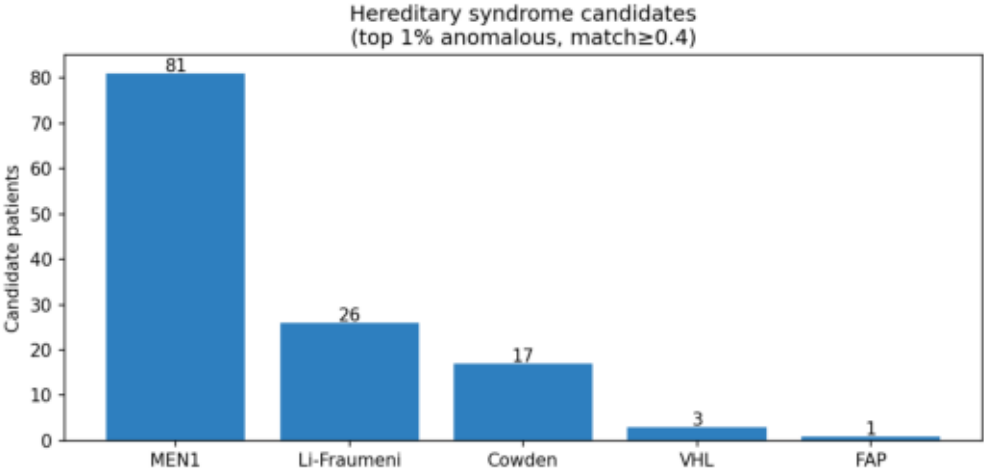
Cowden: ['C50', 'C54']

Autoencoder Anomaly Scores + Syndrome Candidates

Anomaly score distribution (top 1% flagged)

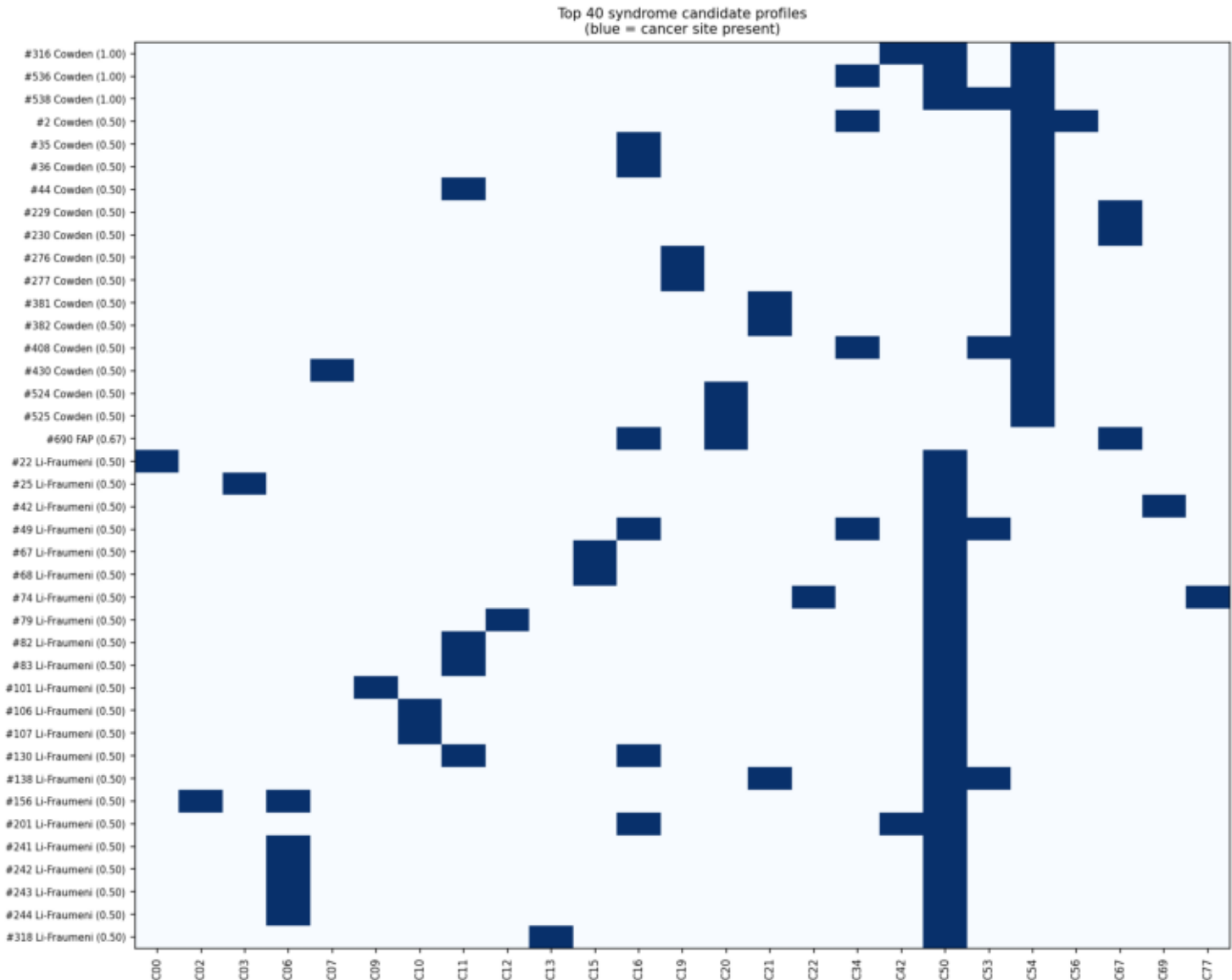


Candidates by syndrome pattern



Syndrome Candidate Profiles

Top 40 candidate profiles — cancer sites present (blue)
ranked by anomaly score; labeled with best-matching syndrome



Candidate Table + Limitations

PID	Rank	N sites	Sites	Best syndrome	Match	Matching sites	Age	Sex
24620731.0	316	3	C43+C58+C51	Fewer	0.0	C50+C54	49.0	F
20179958.0	536	3	C34+C50+C54	Cowden	1.0	C50+C54	46.0	F
20515318.0	538	3	C50+C53+C54	Cowden	1.0	C50+C54	57.0	F
29442914.0	2	3	C34+C54+C56	Cowden	0.5	C54	64.0	F
24181848.0	35	2	C16+C54	Cowden	0.5	C54	61.0	F
16746866.0	36	2	C16+C54	Cowden	0.5	C54	56.0	F
14173041.0	44	2	C11+C54	Cowden	0.5	C54	40.0	F
21283792.0	229	2	C54+C67	Cowden	0.5	C54	72.0	F
28107980.0	230	2	C54+C67	Cowden	0.5	C54	67.0	F
485689.0	276	2	C19+C54	Cowden	0.5	C54	55.0	F
21563992.0	277	2	C19+C54	Cowden	0.5	C54	89.0	F
22276543.0	381	2	C21+C54	Cowden	0.5	C54	53.0	F
897484.0	382	2	C21+C54	Cowden	0.5	C54	63.0	F
25122536.0	408	3	C34+C53+C54	Cowden	0.5	C54	50.0	F
19159993.0	430	2	C07+C54	Cowden	0.5	C54	70.0	F
695513.0	524	2	C20+C54	Cowden	0.5	C54	52.0	F
89045.0	525	2	C20+C54	Cowden	0.5	C54	56.0	F
29894368.0	690	3	C16+C20+C67	FAP	0.667	C16+C20	75.0	M
32396261.0	22	2	C00+C50	Li-Fraumeni	0.5	C50	78.0	F
36812018.0	25	2	C03+C50	Li-Fraumeni	0.5	C50	59.0	F
14735245.0	42	2	C50+C69	Li-Fraumeni	0.5	C50	51.0	F
11844634.0	49	4	C16+C34+C50+C53	Li-Fraumeni	0.5	C50	50.0	F
407567.0	67	2	C15+C50	Li-Fraumeni	0.5	C50	46.0	F
34305502.0	68	2	C15+C50	Li-Fraumeni	0.5	C50	66.0	F
813766.0	74	3	C22+C50+C77	Li-Fraumeni	0.5	C50	68.0	F

- Limitations & caveats:
- ① Syndrome definitions use only the 37 registry sites — pituitary (MEN1), sarcoma (Li-Fraumeni), and renal (VHL) are under-represented or absent; match scores are underestimates.
 - ② High anomaly score reflects deviation from the registry average, not necessarily hereditary etiology — rare sporadic co-occurrences will also appear in the tail.
 - ③ No age-of-onset filter applied — hereditary syndromes present earlier; filtering to age_first < 50 would improve specificity.
 - ④ Registry lacks family history data — confirmation requires genetic counselling referral.
 - ⑤ Match threshold (≥0.40) is arbitrary; clinical follow-up should lower threshold to ≥0.33 (2/6 sites).